

Sri Lanka Institute of Information Technology



Lab Submission
Lab - 05

IT25102546

Fonseka D.V.A.N

Artificial Intelligence and Machine Learning | IT2011

B.Sc. (Hons) in Information Technology

Exercise 3 – Comparison of Reasoning LLMs

3.1 DeepSeek

Mode used: DeepThink/Reasoning

Prompt:

Generate a detailed report on how a university student can use generative AI to learn Probability and Statistics, Database Management Systems, and Software Engineering. For each subject, explain: 1. How generative AI can support learning 2. Practical examples 3. Suitable study activities 4. Benefits 5. Risks and limitations 6. How to use AI ethically without depending on it completely Use clear headings and simple language.

Generated report:

Report: Using Generative AI to Learn University Subjects

Introduction

Generative AI (like ChatGPT, Gemini, or Claude) is a powerful tool for university students. It can explain ideas, create examples, and give feedback. However, it is not a replacement for your own brain. This report shows how to use AI wisely for three subjects: Probability and Statistics, Database Management Systems, and Software Engineering. For each subject, we will look at practical ways to learn, the good points, the dangers, and how to stay honest.

Subject 1: Probability and Statistics

1. How Generative AI Can Support Learning

AI can help you understand difficult concepts like distributions, hypothesis testing, and regression. It can act like a personal tutor that never gets tired. You can ask it to explain a formula in plain words, show step-by-step solutions, or create new practice problems. It can also help you visualize data by writing code for graphs.

2. Practical Examples

- Ask: `“Explain the Central Limit Theorem like I am 15 years old.”`
- Ask: `“Give me 5 real-world examples of where we use Poisson distribution.”`
- Ask: `“Write Python code to plot a normal distribution curve using matplotlib.”`
- Give AI a data set (in text) and ask: `“Calculate the mean, median, and standard deviation, and tell me what they mean.”`

3. Suitable Study Activities

- `“Concept clarification:”` Read your textbook, then ask AI to re-explain any confusing paragraph.

- **Practice problem generation:** Ask AI to create 10 multiple-choice questions on confidence intervals. Solve them, then ask AI to check your answers.
- **Step-by-step walkthrough:** Give AI a problem and ask it to solve it one line at a time. After each line, pause and try the next step yourself before looking.
- **Data storytelling:** Give AI a small dataset and ask it to suggest which statistical test to use, then run the test using AI-generated code.

4. Benefits

- Saves time on repetitive calculations (if you already know the method).
- Gives many different explanations until one “clicks” for you.
- Helps you connect theory to real-life situations.
- Generates instant practice material, so you never run out of problems.

5. Risks and Limitations

- AI often makes calculation errors, especially with complex numbers.
- It may give confident but wrong explanations (hallucinations).
- It can skip important assumptions (e.g., normality, independence) that are critical in statistics.
- Over-relying on AI for calculations stops you from developing number sense.

6. How to Use Ethically Without Depending Completely

- Never ask AI to do your homework assignments directly. Instead, use it to understand the *method*.
- Always double-check AI’s numerical answers with a calculator or manual work.
- For exams, practice without AI at least 70% of the time. Use AI only for initial learning or error checking.
- Keep a notebook where you write down every new concept in your own words after discussing it with AI.

Subject 2: Database Management Systems (DBMS)

1. How Generative AI Can Support Learning

DBMS involves SQL queries, database design (ER diagrams), normalization, and transactions. AI is excellent at writing SQL code, explaining joins, and converting business rules into tables. It can also simulate a database environment where you ask questions and get query answers instantly.

2. Practical Examples

- Ask: **“Write an SQL query to find customers who bought more than 5 items in the last month from tables ‘Orders’ and ‘Customers’.”**

- Ask: **“What is the difference between INNER JOIN and LEFT JOIN? Show with small example tables.”**
- Ask: **“Given these attributes: StudentID, Name, Course, Professor, Grade. Normalize this into 3NF and explain why.”**
- Ask: **“Generate an ER diagram in text format for a library system with books, members, and loans.”**

3. Suitable Study Activities

- **Query writing practice:** Describe a problem in plain English, try to write the SQL yourself, then ask AI to write its version. Compare and discuss differences.
- **Design challenges:** Give AI a scenario (e.g., “online store”). Design tables on paper, then ask AI to critique your design for redundancy or missing keys.
- **Error debugging:** Copy an error message from your SQL client into AI and ask for possible fixes. But try to fix it yourself first.
- **Transaction simulation:** Ask AI to explain ACID properties using a bank transfer example, then ask it to show you the SQL commands inside a transaction.

4. Benefits

- Speeds up learning syntax – you can focus on logic, not memorizing commas.
- Provides instant feedback on your query logic.
- Helps you see many alternative ways to write the same query.
- Makes abstract topics (like normalization) concrete with step-by-step transformations.

5. Risks and Limitations

- AI may generate queries that work but are very slow or inefficient (missing indexes).
- It may not understand your specific database schema correctly.
- It can suggest designs that ignore security or real-world constraints.
- If you copy-paste AI queries without understanding, you will fail practical lab exams.

6. How to Use Ethically Without Depending Completely

- Treat AI as a “query helper,” not a “query writer.” Always explain every line of an AI-generated query in your own words before using it.
- For assignments, submit your own work. You can use AI to check your query, but rewrite it from memory afterward.
- Practice writing SQL on paper without any AI – this is how exams often test you.
- Use AI to generate ***sample data*** to test your own queries, not to give final answers.

Subject 3: Software Engineering

1. How Generative AI Can Support Learning

Software Engineering is about planning, designing, coding, testing, and managing projects. AI can help with code generation, debugging, writing test cases, explaining design patterns, and even suggesting project timelines. It can act as a senior developer who gives advice on architecture and best practices.

2. Practical Examples

- Ask: `*"Write a Python class for a simple banking account with deposit and withdraw methods."*`
- Ask: `*"What is the Model-View-Controller pattern? Give a simple example in Java."*`
- Ask: `*"Generate unit test cases for a function that adds two numbers."*`
- Ask: `*"Review this code (paste code) and suggest improvements for readability and performance."*`
- Ask: `*"Create a basic project plan (with milestones) for building a chat app in 8 weeks."*`

3. Suitable Study Activities

- **Code reading and explanation:** Take a piece of AI-generated code and add detailed comments explaining every block. This builds understanding.
- **Design discussion:** Describe a project requirement to AI and ask it to suggest three different architectures. Then compare them and choose one with reasons.
- **Bug hunting:** Write your own buggy code, give it to AI to find errors, but try to locate them yourself first.
- **Agile simulation:** Ask AI to act as a product owner. You write user stories, and AI gives feedback on clarity and priority.
- **Refactoring practice:** Take old code, ask AI for refactoring suggestions, then manually apply and test the changes.

4. Benefits

- Reduces time spent on boilerplate code, so you can focus on logic and design.
- Gives you exposure to different coding styles and best practices.
- Helps you learn new libraries or frameworks quickly with working examples.
- Builds confidence in debugging and testing through interactive problem-solving.

5. Risks and Limitations

- AI code may contain security vulnerabilities (e.g., SQL injection, weak passwords).
- It often gives “toy” solutions that do not scale for large projects.

- It cannot understand the full context of your project, so its suggestions may be wrong.
- Overuse makes you a poor debugger – you become lost when AI cannot fix a problem.

6. How to Use Ethically Without Depending Completely

- Never submit AI-generated code as your own without understanding it. Most universities consider this plagiarism.
- Use AI for **ideas**, not **solutions**. For example, ask for outline of a sorting algorithm, then code it yourself.
- Before using any AI code, write your own version from scratch. Then compare and learn from the differences.
- For team projects, clearly document which parts you used AI for and which you built yourself – be transparent.
- Spend more time on design, planning, and testing than on coding. AI can help with code, but only you can think about user needs and project goals.

General Ethical Rules for All Subjects

- ****Always cite AI help**** if your professor allows it.
- ****Use AI as a second brain****, not as a replacement for your own thinking.
- ****Set a timer**** – if you use AI for more than 20% of your study session, stop and work alone.
- ****Check with your instructor**** about their policy on AI tools.
- ****Remember:**** The goal is to become an expert. AI is just a tool, like a calculator or a textbook.

Final Summary Table

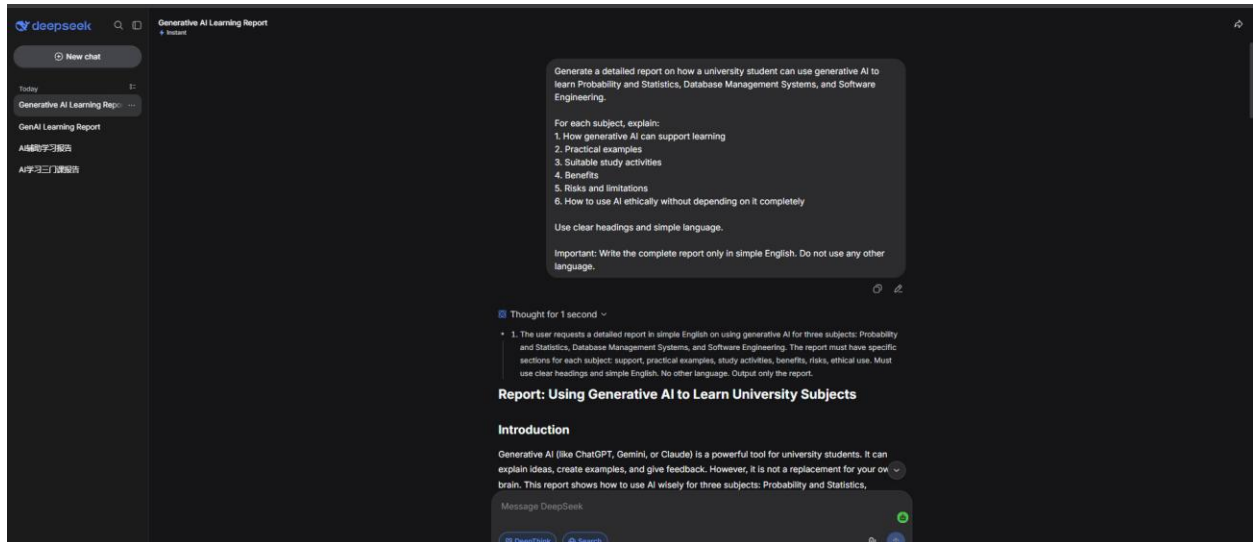
Subject	Best Use of AI	Biggest Danger	Ethical Habit
-----	-----	-----	-----
Probability & Statistics	Explaining concepts and generating practice	Wrong calculations	Double-check numbers manually
Database Management	Writing SQL and designing schemas	Inefficient or unsafe queries	Write queries yourself first
Software Engineering	Generating boilerplate code and test cases	Security holes and poor architecture	Refactor and understand every line

Conclusion

Generative AI is a wonderful study partner for university students. It can make hard topics easier and save valuable time. But it is not a magic answer machine. To truly learn Probability and Statistics, Database Management Systems, and Software Engineering, you must use AI actively – asking questions, checking answers, and always thinking critically. The best students will use AI

to **enhance** their skills, not to **avoid** the hard work. Study smart, stay honest, and always keep learning with your own mind first.

Screenshots



3.2 Qwen

Mode used: Thinking

Prompt:

Generate a detailed report on how a university student can use generative AI to learn Probability and Statistics, Database Management Systems, and Software Engineering. For each subject, explain: 1. How generative AI can support learning 2. Practical examples 3. Suitable study activities 4. Benefits 5. Risks and limitations 6. How to use AI ethically without depending on it completely Use clear headings and simple language.

Generated report:

A Student's Guide to Using Generative AI for Learning

****Subjects:**** Probability and Statistics, Database Management Systems (DBMS), and Software Engineering

Generative AI can be a powerful study companion for university students. When used correctly, it acts as a personalized tutor, a practice partner, and a debugging assistant. Below is a detailed, subject-by-subject report on how to leverage generative AI effectively, along with practical examples, benefits, risks, and ethical guidelines.

1. Probability and Statistics

How Generative AI Can Support Learning

AI can break down complex mathematical concepts into simple, step-by-step explanations. It can generate endless practice problems tailored to your skill level, simulate data distributions, and write code (like Python or R) to help you visualize statistical concepts.

Practical Examples

- **Concept Explanation:** "Explain the Central Limit Theorem to me like I'm a beginner, using a real-world example like rolling dice or measuring human heights."
- **Code Generation:** "Write a short Python script using `matplotlib` to simulate 1,000 coin flips and plot the distribution of heads."
- **Step-by-Step Help:** "I got stuck on this Bayes' Theorem problem. Here is my work so far. Can you tell me where my logic went wrong without giving me the final answer?"

Suitable Study Activities

- **Interactive Quizzing:** Ask the AI to generate a 5-question quiz on hypothesis testing, then grade your answers and explain any mistakes.
- **Formula Flashcards:** Prompt the AI to create a table of key probability distributions, their formulas, and when to use them.
- **Data Simulation:** Use AI to generate small, custom datasets to practice calculating mean, variance, and standard deviation by hand.

Benefits

- Provides immediate, judgment-free feedback.
- Adapts to your learning pace (e.g., asking for simpler or more advanced explanations).
- Bridges the gap between abstract math and practical coding/visualization.

Risks and Limitations

- **Hallucinations:** AI can make subtle mathematical errors or invent fake formulas.
- **Surface Learning:** Copying AI-generated solutions prevents you from developing problem-solving intuition.
- **Calculation Dependency:** Relying on AI to do the math means you might fail during a closed-book exam.

How to Use AI Ethically and Avoid Dependence

- **The "Try First" Rule:** Always attempt the problem on your own before asking the AI for help.
- **Ask for Hints, Not Answers:** Prompt the AI with, "Give me a hint for the next step," rather than "Solve this."
- **Verify with Textbooks:** Cross-check AI-generated formulas and steps with your official course materials.

2. Database Management Systems (DBMS)

How Generative AI Can Support Learning

AI excels at translating natural language into SQL queries, explaining relational database concepts (like normalization and joins), and helping you design database schemas. It can also act as a debugging tool for syntax errors.

Practical Examples

- **Query Translation:** "Write a PostgreSQL query to find the top 3 students with the highest GPA in each department. Then, explain what each part of the query (SELECT, JOIN, GROUP BY) does."
- **Schema Review:** "Here is my table structure for a library system. Are there any violations of the Third Normal Form (3NF)? How can I fix them?"
- **Error Debugging:** "I got this error message: 'column must appear in the GROUP BY clause'. Here is my query. Explain why this error happens and how to fix it."

Suitable Study Activities

- **SQL Practice Sandbox:** Ask the AI to generate a mock dataset (e.g., an e-commerce store) and then give you 5 progressively harder SQL challenges to solve on that data.
- **Role-Playing Interviews:** Prompt the AI: "Act as a senior database administrator. Ask me 3 interview questions about index optimization and evaluate my answers."
- **Diagram Generation:** Ask the AI to generate Mermaid.js code for an Entity-Relationship (ER) diagram based on your text description, which you can then visualize.

Benefits

- Accelerates the learning of complex SQL syntax and dialect differences.
- Helps visualize abstract database designs before building them.
- Provides instant explanations for confusing error messages.

Risks and Limitations

- **Dialect Confusion:** AI might mix up SQL dialects (e.g., giving you MySQL syntax when you are using PostgreSQL).
- **Inefficient Queries:** AI might generate queries that work but are highly inefficient or vulnerable to SQL injection.
- **Illusion of Competence:** Reading a correct SQL query is easy; writing one from scratch under exam conditions is hard.

How to Use AI Ethically and Avoid Dependence

- **Write First, Optimize Later:** Draft your SQL query manually, then ask the AI, "How can I make this query more efficient or readable?"

- ****Test Everything:**** Always run AI-generated queries in a real database environment (like SQLite or PostgreSQL) to verify they work.
- ****Understand the "Why":**** Never use an AI-generated query in your assignment unless you can explain every single clause and keyword out loud.

3. Software Engineering

How Generative AI Can Support Learning

AI can act as a "junior pair programmer." It can explain software design patterns, generate boilerplate code, help write unit tests, review your code for best practices, and clarify Agile/Scrum methodologies.

Practical Examples

- ****Design Patterns:**** "Explain the Model-View-Controller (MVC) pattern using a simple Java or Python example. What are the pros and cons?"
- ****Code Review:**** "Here is a function I wrote. Act as a senior developer and review it for readability, potential bugs, and adherence to SOLID principles."
- ****Test Generation:**** "Generate 5 unit tests for this Python function, including edge cases like empty inputs or negative numbers."

Suitable Study Activities

- ****Refactoring Practice:**** Write a messy, working piece of code, then ask the AI to suggest three different ways to refactor it for better performance or readability.
- ****Documentation Drafting:**** Ask the AI to help you draft a README file or API documentation for your project, then edit it to ensure it accurately reflects your code.
- ****Architecture Brainstorming:**** "I am building a task-management app. What are the pros and cons of using a monolithic architecture versus microservices for this?"

Benefits

- Bridges the gap between theoretical computer science and real-world coding.
- Saves time on repetitive tasks (like writing boilerplate or basic tests), allowing you to focus on system design.
- Improves code quality through instant, detailed feedback.

Risks and Limitations

- ****Outdated Knowledge:**** AI might suggest deprecated libraries or outdated security practices.
- ****Security Flaws:**** AI can inadvertently generate code with security vulnerabilities (e.g., hardcoded credentials, poor input validation).
- ****Plagiarism:**** Submitting AI-generated code as your own violates most university academic integrity policies.

How to Use AI Ethically and Avoid Dependence

- **Treat AI as a Reviewer, Not a Writer:** Write the core logic and architecture yourself. Use AI only to review, test, or refine your work.
- **Line-by-Line Understanding:** If you include AI-suggested code in your project, you must be able to explain every line of it to your professor.
- **Disclose Usage:** Always follow your university's guidelines on AI disclosure. If in doubt, cite the AI tool in your project documentation.

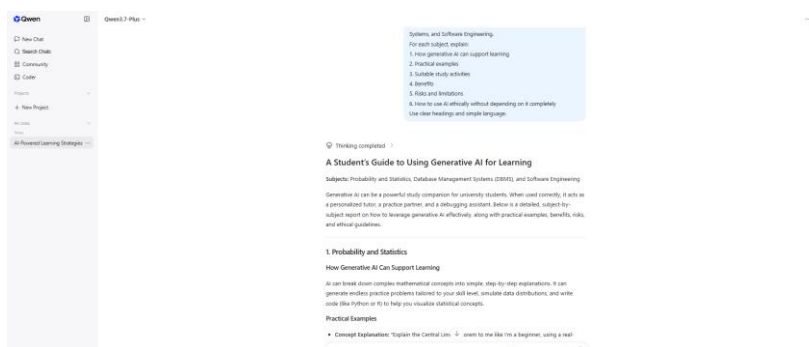
General Guidelines: Using AI Ethically Without Over-Dependence

To ensure generative AI enhances your learning rather than replacing it, adopt these universal habits:

1. **The "Struggle is Good" Rule:** Learning happens when you struggle with a problem. If you immediately ask AI for the answer, you rob yourself of the cognitive effort required to build long-term memory.
2. **Prompt for Process, Not Product:** Instead of asking "What is the answer?", ask "What is the first step I should take to solve this?" or "What concept am I misunderstanding here?"
3. **Triangulate Information:** Never trust a single AI output blindly. Verify facts, formulas, and code against your textbook, lecture notes, or official documentation.
4. **Maintain Academic Integrity:** Know your university's specific policy on generative AI. When in doubt, assume that submitting AI-generated work as your own is a violation of academic honesty.
5. **Turn It Off Occasionally:** Regularly practice studying, coding, and solving problems without any AI assistance to ensure you are truly prepared for exams and real-world scenarios.

By treating generative AI as a **tutor and assistant** rather than a **shortcut**, you can deepen your understanding of Probability, DBMS, and Software Engineering while staying firmly in control of your own education.

Screenshot:



3.3 Grok

Mode used: Fast - DeepSearch mode was unavailable under the free account. Therefore, the available Fast mode was used.

Prompt:Generate a detailed report on how a university student can use generative AI to learn Probability and Statistics, Database Management Systems, and Software Engineering. For each subject, explain:

How generative AI can support learning

Practical examples

Suitable study activities

Benefits

Risks and limitations

How to use AI ethically without depending on it completely Use clear headings and simple language. Important: Write the complete report only in simple English. Do not use any other language.

Generated report:

Using Generative AI to Learn Probability and Statistics, Database Management Systems, and Software Engineering

Generative AI tools such as ChatGPT, Claude, Gemini, or similar systems can help university students learn more effectively. These tools create explanations, examples, practice questions, and feedback based on what a student asks. This report explains how a student can use generative AI for three subjects: Probability and Statistics, Database Management Systems, and Software Engineering. For each subject, the report covers support for learning, practical examples, suitable study activities, benefits, risks and limitations, and ethical use without full dependence on AI.

The advice stays simple and practical. Students should always check AI answers against textbooks, lecture notes, and reliable sources.

Probability and Statistics

1. How Generative AI Can Support Learning

Generative AI can break down hard ideas into easy steps. It can explain formulas in plain words, show how concepts connect, and create practice problems with solutions. A student can ask for different ways to understand the same topic until it becomes clear. AI can also check a student's own calculations or explanations and point out mistakes.

2. Practical Examples

- Ask AI to explain the difference between probability and statistics with everyday examples such as coin tosses or exam scores.

- Request a step-by-step solution for finding the mean, median, and standard deviation of a small data set.

- Ask for a simple story that shows how Bayes' theorem works in medical testing.
- Have AI generate ten practice questions on normal distribution and then check the student's answers.

3. Suitable Study Activities

- Turn lecture notes into short summaries and then expand them with extra examples from AI.
- Create flashcards by asking AI for key terms and definitions, then test yourself without looking at the answers.
- Solve textbook problems first, then ask AI to review the work and explain any errors.
- Practice explaining a concept to AI as if teaching a friend, and request feedback on clarity.
- Generate random data sets and calculate summary statistics by hand before verifying with AI.

4. Benefits

AI makes abstract ideas easier to grasp through many examples. It gives instant practice material and helps students see common mistakes. Learning becomes more active because students can explore topics at their own pace. Confidence grows when difficult sections become clearer.

5. Risks and Limitations

AI can give wrong formulas or incomplete explanations. It may invent data or skip important conditions. Students who only copy answers learn less. Over-reliance can weaken the ability to solve problems independently during exams. AI does not replace real understanding of proofs or assumptions behind methods.

6. How to Use AI Ethically Without Depending on It Completely

Always try to solve problems yourself first. Use AI only for hints, alternative explanations, or checking work. Compare every AI answer with the course textbook and lecture slides. Write notes in your own words after reading AI output. Never submit AI-generated solutions as your own work. Treat AI as a tutor that guides, not a replacement for thinking and practice.

Database Management Systems

1. How Generative AI Can Support Learning

Generative AI can explain database concepts such as tables, relationships, and queries in simple language. It can show how to write SQL statements, design simple schemas, and understand normalization. Students can ask for comparisons between different database types or for help debugging query errors. AI also helps turn theory into practical design steps.

2. Practical Examples

- Ask AI to explain primary keys and foreign keys using a library book system example.
- Request a sample SQL query that selects student names and grades from two related tables.
- Have AI create a simple entity-relationship diagram description for a university course

registration system.

- Ask for step-by-step help in normalizing a messy table into third normal form.
- Request common mistakes in JOIN operations and how to avoid them.

3. Suitable Study Activities

- Write SQL queries by hand, then ask AI to check them and suggest improvements.
- Design a small database schema for a real-life situation such as a sports club, then refine it with AI feedback.
- Convert lecture notes on ACID properties into short questions and answer them before checking with AI.
- Practice translating English requirements into SQL and verify the results.
- Generate practice exercises on indexing or transactions and solve them independently first.

4. Benefits

Students gain clearer mental models of how data is stored and retrieved. Practice material is always available. Error messages become easier to understand. Design skills improve through repeated feedback. Learning feels more practical and less abstract.

5. Risks and Limitations

AI may produce SQL that works only in one database system and fails in another. Schema designs can miss important constraints or security rules. Students who copy queries without understanding them struggle in exams or real projects. AI explanations sometimes oversimplify concurrency or recovery concepts.

6. How to Use AI Ethically Without Depending on It Completely

Design and write queries yourself before asking for help. Use AI to explain why a query failed or to suggest better approaches. Always test queries in a real database environment when possible. Keep personal notes that record what you learned from each interaction. Cite AI assistance if required by university rules. Focus on understanding the principles so you can work without AI later.

Software Engineering

1. How Generative AI Can Support Learning

Generative AI can explain the stages of software development, design patterns, testing methods, and project management ideas in clear language. It can generate sample code, review student code for style and bugs, and suggest better structures. Students can explore how requirements become designs and how designs become working software. AI also helps with documentation and version control concepts.

2. Practical Examples

- Ask AI to explain the difference between waterfall and agile methods with a simple mobile app project.
- Request a basic class diagram description for a library management system.
- Have AI review a short piece of student code and suggest improvements for readability.
- Ask for sample unit test cases for a simple calculator function.
- Request a short explanation of how Git branching works in a team project.

3. Suitable Study Activities

- Write a small program or design first, then ask AI for feedback on structure and style.
- Break a large project requirement into smaller tasks with AI help, then implement the tasks yourself.
- Create practice questions on software life-cycle stages and answer them before checking.
- Generate sample user stories and turn them into simple designs independently.
- Review AI-suggested code improvements and decide which ones to accept based on your knowledge.

4. Benefits

Students see concrete examples of abstract ideas such as modularity and testing. Feedback arrives quickly and helps improve coding habits. Project planning becomes clearer. Documentation skills improve. Learning feels connected to real software work.

5. Risks and Limitations

AI-generated code may contain hidden bugs, security problems, or outdated practices. Designs can miss important non-functional requirements such as performance or security. Students who rely only on AI may not develop strong problem-solving skills. Some AI suggestions ignore course-specific standards or tools.

6. How to Use AI Ethically Without Depending on It Completely

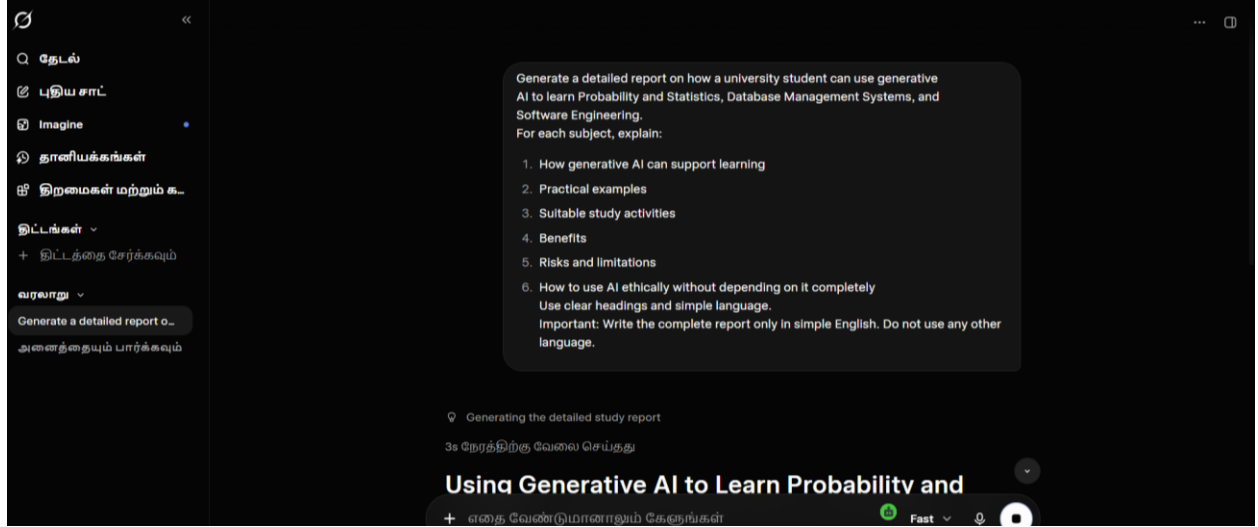
Always write your own code and designs first. Use AI for review, alternative ideas, or explanations of concepts. Test every suggestion carefully. Keep a record of what you changed and why. Follow university rules on AI use and never claim AI work as fully your own. Build the habit of thinking through problems so that AI becomes a helper rather than a crutch.

General Advice for All Three Subjects

Start every study session by reviewing lecture material and attempting problems without AI. Use generative AI to clarify confusion, create extra practice, or receive feedback. Always verify answers with trusted sources. Write summaries and solutions in your own words. Practice explaining ideas out loud or on paper. Over time, reduce the amount of AI help so that your own

skills grow stronger. Ethical use means honesty, independent effort, and respect for academic rules. When used wisely, generative AI becomes a powerful study partner that supports deeper learning without replacing the student's own work.

Screenshots:



- DeepSeek produced a well-structured report with detailed explanations.
- Qwen provided clear and practical study suggestions.
- Grok provided useful current information and wider examples.
- In my opinion, DeepSeek produced the best response because its report was clear, detailed, well-organized, and relevant to all three subjects.